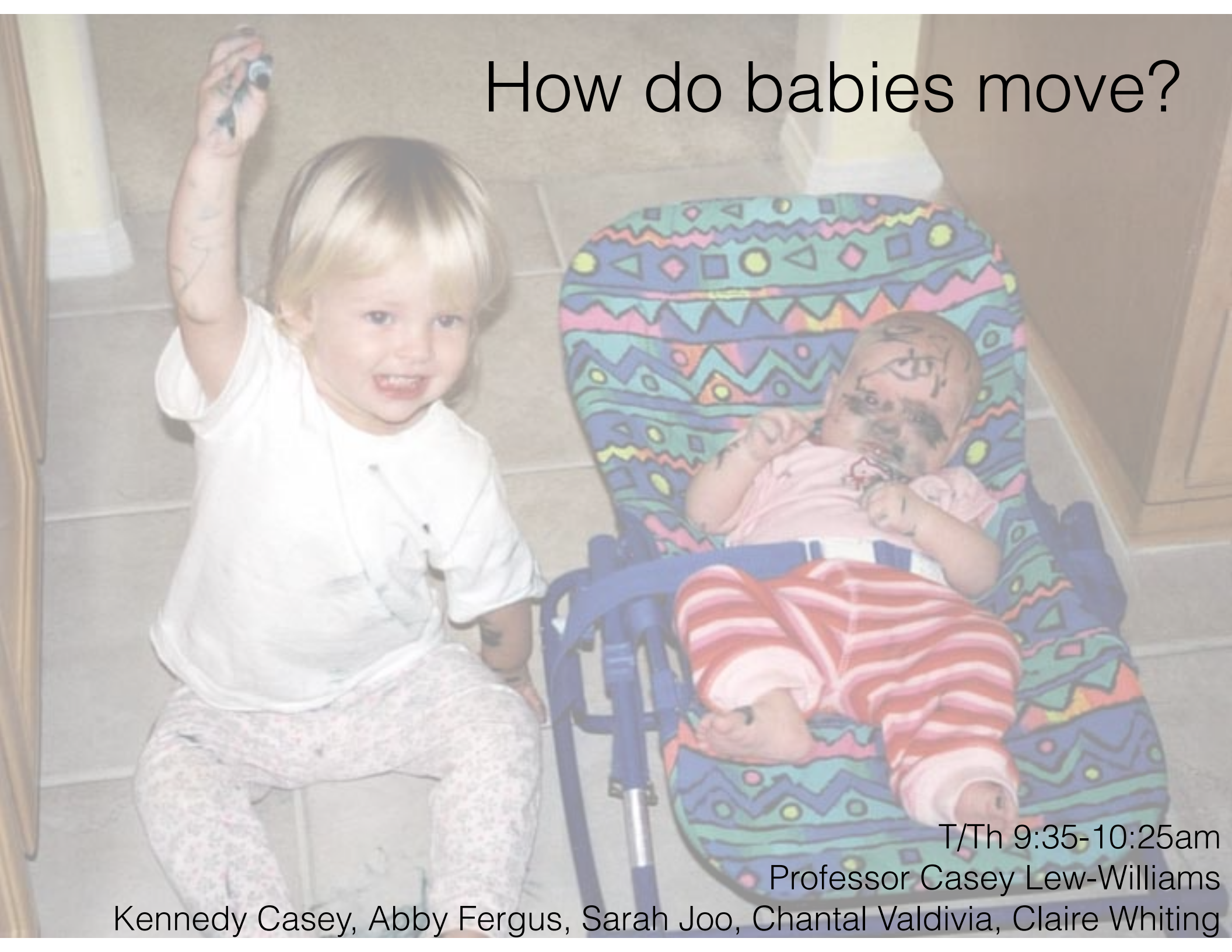


How do babies move?

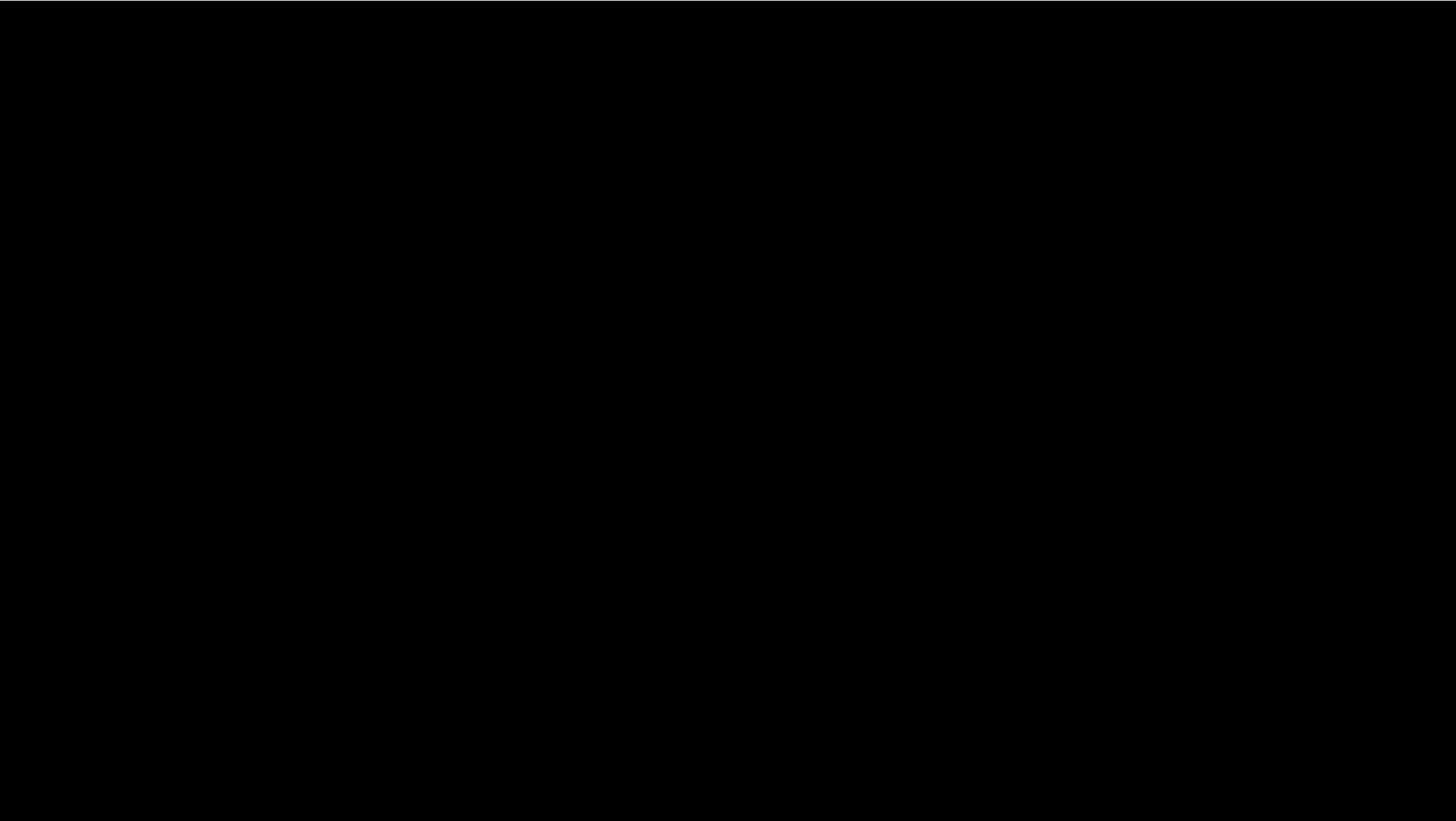


T/Th 9:35-10:25am

Professor Casey Lew-Williams

Kennedy Casey, Abby Fergus, Sarah Joo, Chantal Valdivia, Claire Whiting

Developmental science cares deeply
about understanding change





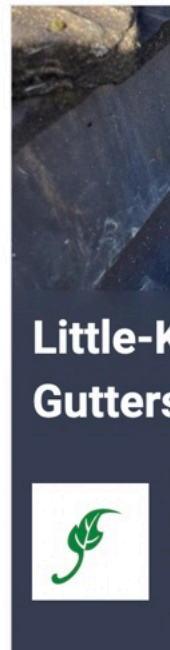
Alpine ibex



True talent

Study: Humans Display Highest Cognitive Abilities When Trying To Retrieve Object Dropped Between Car Seats

Published March 24, 2014



Parkour

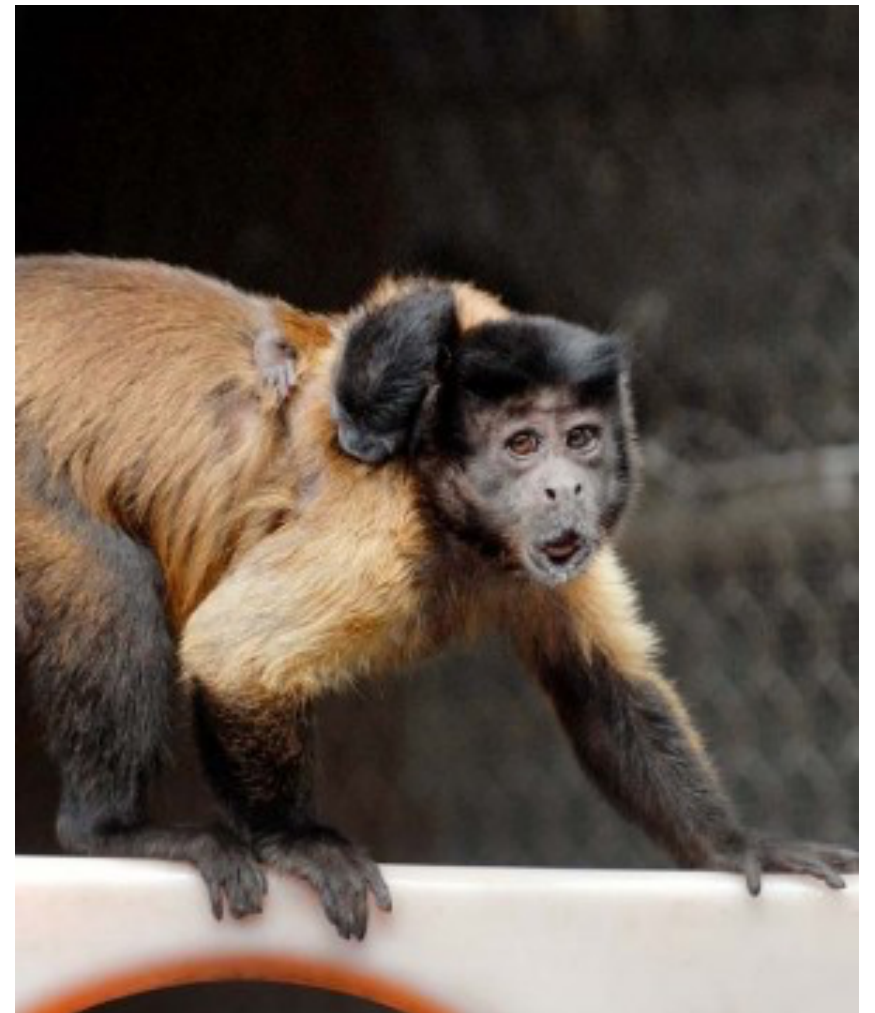


Reflexes: Motor building blocks

- Innate, fixed patterns of action that occur in response to particular stimulation. What is their adaptive value?



Reflexes: Motor building blocks

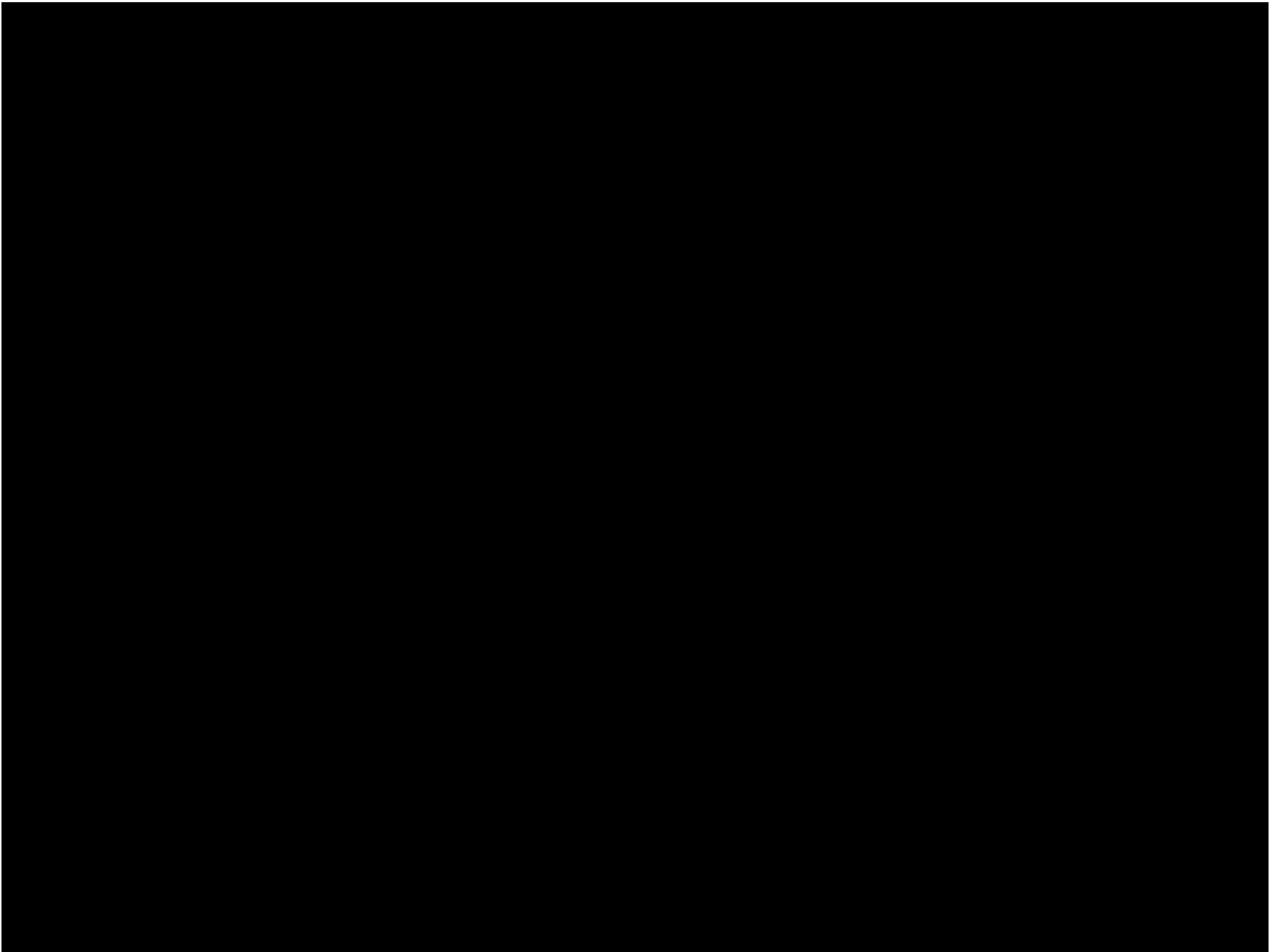




Examples of reflexes

- **Rooting:** stroke cheek —> turn head and open mouth
- **Sucking:** object in mouth —> suck
- **Stepping:** hold a baby upright —> steps
- **Babinski:** stroke toes —> toes fan out
- **Grasping:** object in hand —> grasp
- **Moro:** loud noise or falling —> arms out and then in

Reflexes aren't necessarily motoric



Two types of motor skills

- Fine-motor skills: small muscle movements
- Gross-motor skills: large muscle control



Two types of motor skills



(Chen, Keen, Rosander, & Von Hofsten, 2010)

Motor milestones

Motor skill

Head erect, steady

Prone, lifts self by arms

Rolls from side to back

Grasps cube

Rolls from back to side

Sits alone

Crawls

Pulls to stand

Walks alone

Builds tower of 2 cubes

Jumps in place

90% range

3 weeks - 4 months

3 weeks - 4 months

3 weeks - 5 months

2 - 7 months

2 - 7 months

5 - 9 months

5 - 11 months

5 - 12 months

9 - 17 months

10 - 19 months

17 - 30 months



Motor milestones

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Maturation hypothesis

- Development is based on genetic blueprint, same across most/all humans
- Motor skills are tied to neurological development: maturation occurs in brain regions that control voluntary movements.
- Experience maintains and facilitates development; not central to development, but not irrelevant.

Johnny and Jimmy



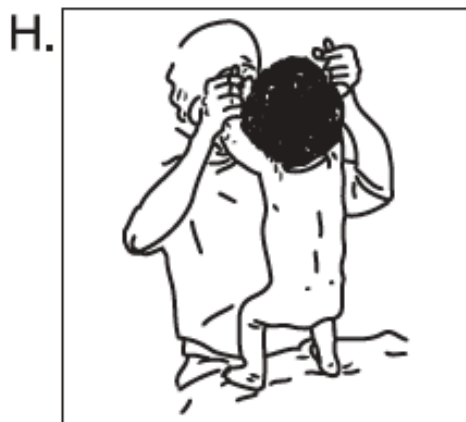
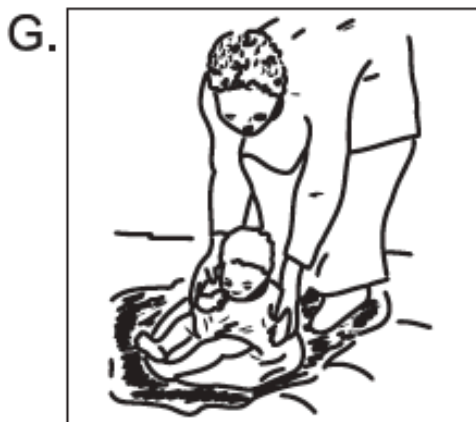
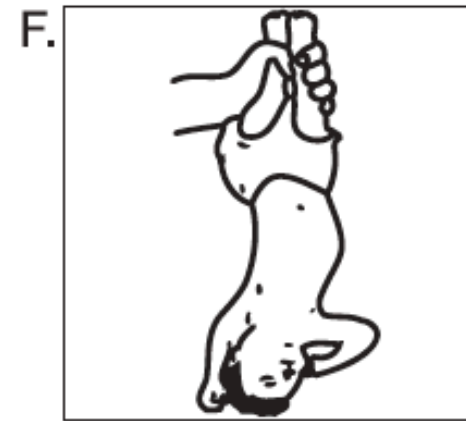
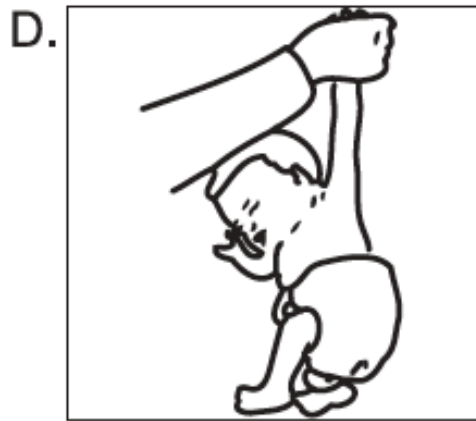
Adverse experience

- Eastern European orphans
- Lack of experience moving —> motor delays

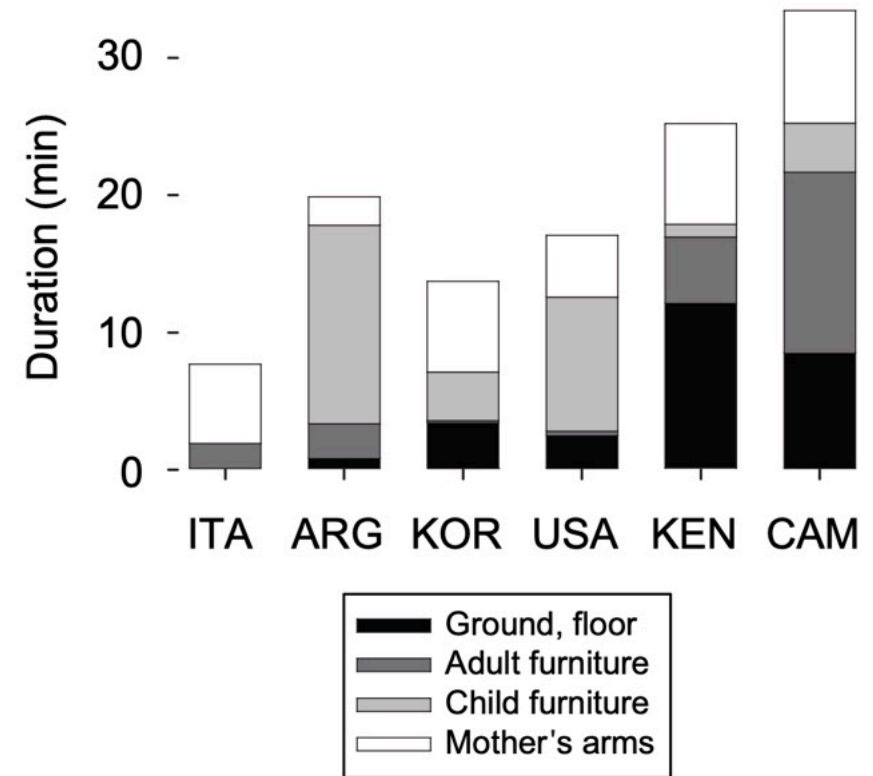
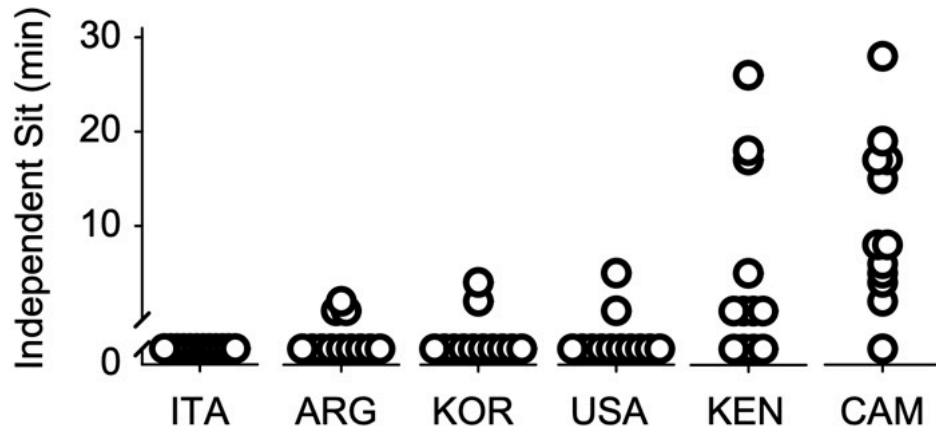


Enhanced experience

Motor development routines practiced in India, and in various African and Caribbean countries



Enhanced experience across cultures



Dynamic systems hypothesis

- (vs. maturational hypothesis)
- This theory considers the developing infant as a complex system composed of many interacting components embedded in a complex and changing environment.
- There's a consistent order in motor skills occurs because abilities interact and build across systems (motor, perceptual, social...)
- Milestones achieved as multiple systems come together.
- Powerful explanation for growth and change - well beyond the motor system.

Stepping reflex



Reaching: Velcro mittens



Crawling



Crawling

- There is no 'program' for crawling assembled in the genes or wired in the nervous system.
- Crawling self-organizes as a solution to a problem (to move across the room), later replaced by more efficient solutions.

Cruising and walking

- Cruising: walking while holding onto objects
- Early walking: disorganized, jerky, recover balance with each step
- Component skills for walking: balance, stepping, interpreting environment, motivation, nervous system, body proportions...
- Re-learning: balance needed for sitting, crawling, walking, and biking (and parkour) does not have the same demands



Transition from sitting to crawling



Transition from crawling to cruising/walking



First steps:)



Creative solutions



Locomotion ↔ Cognition

- Crawling & emotion
 - With crawling, more anger and more affection from the infant
- Crawling/walking & the language environment
 - Infants who spend more time in motion cause caregivers to use more locomotion verbs (*come, bring, walk*)
- Crawling/walking & spatial skills
 - Experienced crawlers are good at using landmarks to find a hidden toy
 - But they fail to use landmarks when they start walking!

Locomotion $\longleftrightarrow$ Cognition

